PWLLHELI SAILING CLUB

Race Hut System Costs

The start hut, the yacht trackers and the services that keep them running

Prices as of 6 October 2026.

Equipping the race hut and the yacht trackers cost £5,036 (£4,126 hut, £910 trackers). Running it all costs about £400 a year (£34 a month), plus the Virtual Race Officer's ongoing model cost, which is not yet known.

Summary

Power is nearly half the one-off spend: £2,348 of £5,036. Prices are what was paid, including VAT and delivery.

Area	One-off spend	Bought
Hut: power (Victron kit, 400 Ah battery, 2 solar panels, fusing, cabling)	£2,348.11	Jul 2026
Hut: PC, monitor, VE.Direct cables, USB hub	£683.04	Jun–Jul 2026
Hut: horn and horn control box	£569.86	May 2024 – Aug 2026
Hut: 4G router and camera	£415.19	Jun–Jul 2026
Hut: weather station	£109.99	Jun 2026
Yacht trackers, chargers, SIMs, import charges	£909.59	Jul–Aug 2026
Total	**£5,035.78**	

Running costs come to about £403 a year: SMARTY hut data £216, Fasthosts relay £104.40, Hologram tracker SIMs about £75, domain about £8. The VRO's ongoing model cost comes on top and is not yet known.

Hut hardware

The hut kit cost £4,126.19, including a spare horn. That is every item on the hut wiring diagram except the existing club kit listed under Not included.

Area	Item	Supplier	Ordered	Paid
Power	12 V 400 Ah lithium battery (AL12400)	Sterling Power	6 Jul 2026	£840.00
Power	Victron Phoenix Smart IP43 12/30 charger + mains lead, Orion-Tr 12/12-9 isolated DC-DC, SmartShunt 300A, 2 × Blue Sea 5025 fuse blocks, 2 toggle switches	12 Volt Planet	6 Jul 2026	£521.59
Power	Victron SmartSolar MPPT 150/85 VE.Can, solar extension cables, PV DC isolator	12 Volt Planet	8 Jul 2026	£457.18
Power	2 × JA Solar 465 W all-black panels + Fastensol rails, clamps, screws, seals	City Plumbing	8 Jul 2026	£299.89
Power	Blue Sea 2151 dual MRBF terminal fuse block + 2 × Blue Sea 5175 fuses	Amazon	6 Jul 2026	£85.16
Power	2 × 70 A 10 mm² cable + battery disconnect switch	Amazon	7 Jul 2026	£49.89
Power	2 × 3 m Victron solar extension cables	12 Volt Planet	9 Jul 2026	£33.51
Power	2 × Blue Sea 5182 fuses	Amazon	8 Jul 2026	£31.92
Power	MKGT 16 mm² battery cable	Amazon	9 Jul 2026	£28.97

Area	Item	Supplier	Ordered	Paid
PC	MINIX fanless mini PC, Windows 11	Amazon	21 Jun 2026	£452.10
PC	KTC 27-inch 1440p monitor	Amazon	21 Jun 2026	£99.99
PC	3 × Victron VE.Direct-to-USB cables	Amazon	6 & 8 Jul 2026	£99.97
PC	7-port USB 3.0 hub + dual USB 12 V socket	Amazon	7 Jul 2026	£30.98
Horn	Electric air horn, single trumpet, 123 dB, 12 V	ASAP Supplies	5 Jul 2026	£213.16
Horn	Spare air horn, Rupse 150 dB	Amazon	4 Aug 2026	£23.13
Horn	Control box: Delta 240 W 12 V PSU, Omron relay + socket, push button, SP29 connectors, 2 Modulostar fuse holders	RS Components	19 Jan 2025	£157.19
Horn	Control box: enclosure, DIN rail, fuses, fuse holder, early Rupse horn	Amazon	20 Jan 2025	£97.19
Horn	Control box: Tempatron relay + socket, Omron relay + socket, D-sub parts, push button	RS Components	22 May 2024	£79.19
Network	Teltonika RUTX11 4G router	eBay	21 Jun 2026	£219.25
Network	Teltonika surface mounting kit	Amazon	7 Jul 2026	£13.79
Network	CAT5e cable	Amazon	25 Jun 2026	£5.59
Camera	Hikvision DS-2CD3786G2T-IZSY 8 MP dome	eBay	20 Jun 2026	£176.56
Weather	Ecowitt weather station array + gateway	Amazon	21 Jun 2026	£109.99

The two RS orders showed ex-VAT totals (£130.99 and £65.99), so 20% VAT is added above.

Yacht trackers

The trackers cost £909.59. The fleet in use is 5 Teltonika ATC700 and 2 Queclink GL521MG, on the Hologram SIMs. The LL301, GL530MG, FMP100 and AT1 proved unreliable and are no longer used. The White Label SIMs were never used. How each model behaves is in `TRACKERS.md`.

Item	Supplier	Ordered	Paid
4 × Teltonika ATC700	White Label GPS	19 Aug 2026	£259.50
Test set: 1 each of ATC700, LL301, FMP100 and AT1, IP68 magnetic case, 4 multi-network SIMs	White Label GPS	5 Aug 2026	£216.29
Special Delivery upgrade for that order	White Label GPS	5 Aug 2026	£7.88
2 × Queclink GL521MG	Mirifica (Germany)	29 Jul 2026	£167.91
Import VAT £33.42 + handling £12, probably on the Mirifica parcel	Royal Mail	3 Aug 2026	£45.42
GL521MG wireless charger kit, data, UART and USB cables	Mirifica (Germany)	30 Jul 2026	£52.70
3 × Queclink GL530MG	eBay	27 Jul 2026	£102.15
10 × Hologram Hyper eUICC IoT SIMs	Hologram	2 Aug 2026	£57.74

Expanding the tracker fleet

Each extra ATC700 costs about £62 plus delivery, and about £11 a year to run, or about £6 if its SIM is paused over winter.

The figures use the August 2026 prices. An ATC700 from White Label GPS is £52 + VAT (£62.40), and delivery is £9.90 per order. The 3 spare Hologram SIMs cover the first 3 trackers; after that, a 10-pack of SIMs cost £57.74 delivered. Running costs are $1 per SIM a month plus about 7 MB of data ($0.21), the fleet's August average per

SIM: about £0.91 a month per tracker.

Trackers added	Trackers + delivery	SIMs	One-off total	Running, all year	Running, Apr–Oct only
3	£197.10	Spares on hand	£197.10	about £33	about £19
5	£321.90	New 10-pack, £57.74	£379.64	about £55	about £32
10	£633.90	New 10-pack, £57.74	£691.64	about £109	about £64

Traccar, the relay and the hut link should take more boats at no extra cost. The ATC700 charges over USB-C (chargers not costed) and runs about nine hours on a charge, so long and night races need the GL521MGs or an ATC700 on a USB-C power bank (with its power bank setting switched on). Slowing the reporting rate only stretches the battery usefully at every 5 minutes or longer (1.1–1.7% an hour, measured parked), and fixes that far apart are too coarse to detect a finish.

Running costs

The known services cost about £403 a year, or £34 a month. The hut data SIM is over half of that. Dollar amounts are converted at about $1.33 to the pound.

Service	What it is	Per month	Per year
Hut 4G data	SMARTY Unlimited on the line named "Hut". £20 list, 10% discount, renews on the 30th	£18.00	£216.00
Relay server	Fasthosts VPS 4, 12-month contract to 29 Jul 2027. Then £99 + VAT	£8.70	£104.40 (£118.80 from Aug 2027)
Tracker SIMs	Hologram Global G3: $1 per SIM a month + $0.03 per MB. 7 SIMs, about 11.5 MB a week across the fleet	about £6.30	about £75 (less if SIMs are paused over winter)
Domain	pwllhelisailingclub.org at Cloudflare Registrar. $30.90 paid covers to 21 Jun 2029	—	about £8 after 2029
Cloudflare account	Tunnels, Access, R2 video and backup storage, caching. Invoices were $0.00–$0.72 a month. Cache Reserve ends 22 Oct 2026	about £0	about £0
Virtual Race Officer	Claude Sonnet through Cloudflare AI Gateway. The $31.50 bought in August went on development and testing	not yet known	not yet known
Electricity	The hut is off-grid: a 24 W median load, met by solar. The mains charger ran in 226 of 122,546 readings	£0	£0

Free services: Open-Meteo forecasts, Mapbox (within its free tier), Traccar (self-hosted on the relay), Victron VRM and MaxMind GeoLite2.

Software

The software costs nothing to license. The race officer app and everything around it are published and documented on GitHub at github.com/mojito9047/pwllheli-race-officer, released under the MIT licence so anyone can use them, and every part they rely on is free.

Part	Software	Licence
Race officer app (hut PC)	Python with Flask, Waitress, pySerial, pyttsx3, Pillow, NumPy, Markdown and pyzipper; FFmpeg for video; Leaflet for the charts	Free and open source
Typefaces	Archivo, Archivo Narrow, IBM Plex Mono	SIL Open Font Licence
Relay	Debian, Caddy, cloudflared, MediaMTX, Traccar, GoAccess	Free and open source
3D films (render machine)	Blender, PyAV, tifffile, fonttools	Free and open source

Part	Software	Licence
Map and weather data	OpenStreetMap and OpenSeaMap charts, Open-Meteo forecasts, Copernicus terrain, Mapbox imagery, MaxMind GeoLite2	Free with attribution: Open-Meteo for non-commercial use, Mapbox within its free tier

The only paid software is Windows, which came with the hut PC and is in its price. The Virtual Race Officer's model use is a running cost, not a licence, and building the app with Claude Code is covered under Development with Claude.

Reducing running costs

Seasonal changes to the two SIM plans save money without losing anything during the season. The relay server can be cut further when its contract renews in July 2027.

Cost	What to do	Saving
Hut 4G SIM, £216 a year	Out of season (Nov–Mar), move the SMARTY line to a small-data plan instead of cancelling, so power monitoring and backups keep running. SMARTY is rolling monthly, with no contract. On quiet days the hut uses about 0.25 GB, so the plan needs about 8 GB a month.	Up to £90 a year, less the smaller plan's cost
Hut 4G SIM	In season, keep Unlimited. The RUTX11 measured 34 GB in September (27 GB sent, 7.5 GB received): about 0.25 GB on quiet days and 1–2 GB on busy ones.	None: Unlimited fits the use
Live stream	Measured on the relay at 1.4 Mbit/s from the camera's 1280×720, 10 fps sub-stream: about 640 MB for each hour someone watches. It ran 82 minutes on 3 Oct (from 11:38) and 24 minutes on 4 Oct, about a quarter of the hut's upload on both days. Setting the sub-stream to 704×576 at 512–768 kbit/s would roughly halve it.	Halves the stream's data; no cash saving while on Unlimited
Hut 4G SIM	If the clubhouse broadband can reach the hut over a point-to-point Wi-Fi link, the SIM can go. This depends on line of sight, and the link hardware is a one-off cost.	Up to £216 a year
Relay server, £104.40 a year	Before the July 2027 renewal, compare Fasthosts VPS 4 with what the relay needs (about 1 vCPU, 512 MB–1 GB RAM, 4 GB disk), then move to a smaller plan or a cheaper provider. Keep it off a member's home server.	Depends on the plan chosen
Tracker SIMs, about £75 a year	Pause the 7 Hologram SIMs over winter: the $1 per SIM monthly fee is about 85% of the cost. Check that paused SIMs are not charged the fee. Keep the 3 spare SIMs unactivated.	About £25 a year
Cloudflare	Let Cache Reserve end on 22 Oct as scheduled. Add an R2 lifecycle rule for past seasons' public clips (about 3.5 GB a year) so storage stays inside the free allowance.	Keeps it at about £0
Teltonika RMS	Let the credit lapse on 24 Oct without renewing.	Keeps it at £0

The VRO's ongoing cost should be small. The test runs put it at about 1.6 US cents per sentence before caching, on Sonnet 5. That is under $1 on a busy race day, and perhaps $20–50 a season. Prepaid AI Gateway credits cap it, so top up in small amounts.

Development with Claude

Building the app with Claude Code used about 8.1 billion tokens from 28 July to 6 October 2026. At pay-as-you-go API prices that would be about $5,170 (about £3,900). It was paid for by subscription instead, not per token.

Model	Replies	Output tokens	Cache reads	Cache writes	API-equivalent
Claude Opus 5	14,679	10.7 M	7,341 M	97.1 M	$4,909

Model	Replies	Output tokens	Cache reads	Cache writes	API-equivalent
Claude Opus 5.5	1,288	1.1 M	522 M	6.6 M	$178
Claude Fable 5.1	245	0.4 M	82 M	1.2 M	$63
Claude Opus 4.8	46	0.03 M	26 M	0.6 M	$20
Total	**16,258**	**12.3 M**	**7,972 M**	**105.4 M**	**$5,170**

Nearly all the tokens are cache reads: each reply re-reads the session so far, at a tenth of the fresh-input price or less. August was the busiest month, at $3,006 of the total.

The count covers only the session transcripts kept on the development PC, which start on 28 July. Earlier development is not included, so the true token total is higher.

Not included

The only cost still open is the VRO's ongoing model use.

Existing club kit, so not costed: the VHF handheld and its 12 V cradle, the USB–RS232 adapter for the horn box, and the keyboard, mouse and desk speaker.

Sources: order and invoice emails. The equipment list comes from the hut wiring diagram v0.2 and the horn control box drawing. The tracker and power figures come from the trackers guide and the hut's power history.